

TEMPLATE

KEY PROJECT INFORMATION & PROJECT DESIGN DOCUMENT (PDD)

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VERSION **v. 1.2**

RELATED SUPPORT

- TEMPLATE GUIDE Key Project Information & Project Design Document v.1.2

This document contains the following Sections

Key Project Information

SECTION A – Description of project

SECTION B - Application of approved Gold Standard Methodology (ies) and/or demonstration of SDG Contributions

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Appendix 3 - Summary of Approved Design Changes (project specific)

KEY PROJECT INFORMATION

GS ID of Project	GS11427
Title of Project	Installation of Household Biogas plants in various states of India
Time of First Submission Date	11/11/2021
Date of Design Certification	N/A
Version number of the PDD	4
Completion date of version	06/07/2022
Project Developer	Greneity Infocom Service Private Limited
Project Representative	Greneity Infocom Service Private Limited
Project Participants and any communities involved	Greneity Infocom Service Private Limited Navrang Seva Sansthan
Host Country (ies)	India
Activity Requirements applied	☒ Community Services Activities ☐ Renewable Energy Activities ☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Scale of the project activity	☐ Micro scale ☒ Small Scale ☐ Large Scale
Other Requirements applied	N/A
Methodology (ies) applied and version number	AMS-I.E. Switch from non-renewable biomass for thermal applications by the user - Version 12
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A
Project Cycle:	☐ Regular ☒ Retroactive

Table 1 – Estimated Sustainable Development Contributions

Sustainable Development Goals Targeted	SDG Impact (defined in B.6.)	Estimated Annual Average	Units or Products
13 Climate Action (mandatory)	Emission reductions	46,657	tCO ₂ e
7 Affordable and Clean Energy	Access to affordable and clean energy services	10,702 biogas plant users are expected to access clean energy	Number of affected users
8 Decent Work and Economic Growth	Unemployment rate, by sex, age and persons with disabilities	15 permanent employments, excluding more than 20 contractual employments.	Number of employment created.
8 Decent Work and Economic Growth	Unemployment rate, by sex, age and persons with disabilities	2 trainings per year.	Training will be provided
3 Good health and well being	Improvement in health and decrease in illness	10,702 biogas plant users will have improved health conditions.	Number of affected users.

SECTION A. DESCRIPTION OF PROJECT

A.1 Purpose and general description of project

>> The project activity aims to replace the commonly used inefficient wood fired mud stove technology, with efficient biogas based cook stove, which is clean and sustainable. The project activity involves bundling household biogas plants located in the states of Odisha, Punjab, West Bengal, Bihar and Jharkhand, India with varying capacities – 2m³, 3m³, 4m³ and 6m³. The proposed project activity includes 10,702 digesters which were implemented between 15 November, 2020 and 22nd August, 2021.

Each household will utilize the dung of its cows to feed the digester for the production of biogas for domestic purposes. The generated biogas from the plants to be used for the purpose of cooking and other thermal energy needs in households. The thermal energy generated from the project activity replaces the equal amount of thermal energy which otherwise would have been supplied from the woody biomass (fire-wood) based stove technology. Therefore, the project activity replaces firewood for equal thermal energy needs and thereby reduces harmful

greenhouse gases in the atmosphere. In addition, the hygienic conditions in the rural areas will be improved by an appropriate disposal of waste. Further, residue from the bio digesters is used as organic fertilizer and will improve soil conditions in rural areas.

Pre project Scenario:

Household survey was conducted to assess the baseline fuel and quantity used. As per the Survey, firewood was the main fuel used to suffice domestic energy needs (cooking). It was sourced from nearby forests and open market. Usage of firewood in inefficient cooking system (conventional system) leads to indoor pollution and land use patterns have been showing a decrease in forest land cover and increase in degraded land. A trend of forests turning into open scrubs has been observed. Degradation of forest lands has exacerbated the already existing problem of desertification. There is a need to maintain adequate forest cover in the state to mitigate climate change effects.

Project Scenario:

The project activity involves bundling of 10,702 household biogas plants located in the states of Odisha, Punjab, West Bengal, Bihar and Jharkhand, India with varying capacities – 2m³, 3 m³, 4 m³ and 6m³. The project development is performed & maintained by Navrang Seva Sansthan and its supported NGOs. Project activity involves installation of new biogas plant in the households of Odisha, Punjab, West Bengal, Bihar and Jharkhand, and produced biogas will be used in the biogas stoves for thermal energy needs. Hence, the project activity is a Greenfield project activity. The size of the bio digesters varies, depending on the number of people and number of cattle available per household.

Residues from the bio digesters are used as organic fertilizer in the garden area and fields of the local people.

Project activity will result in average saving of 46,657 tCO₂e/year.

A.1.1. Eligibility of the project under Gold Standard

>> The project is eligibility under GS4GG as the project activity involves dissemination of household biogas which is eligible under Gold Standard pursuant to clause 3.1.1 of the GS4GG Principles and Requirements document. The eligibility compliance under GS4GG is detailed as follows;

Eligibility Criteria Category	Eligibility criterion - Required condition	Justification
1. Types of Project	Eligible projects shall include physical action/implementation on	The project is already implemented since its start date.

	the ground. Pre-identified eligible project types are identified in the Eligibility Principles and Requirements section.	As per section 3.1.1 (a) of GG4GG Community Services Activity Requirements ver.1.2," Renewable energy types such as solar (photovoltaic and solar thermal electricity generation), tidal/wave, wind, hydropower, geothermal, waste to energy and renewable biomass that are connected to mini grid3 or off grid solutions for targeted users and/or applications." Since, project activity includes utilization of cow dung as feed to bio-digesters along with other organic waste to generate biogas Hence, it meets the criterion. Thus, the project technology/measure falls under GG4GG Community Services Activity Requirements. Section 4.1.3 of the GS4GG Principles and Requirements document states following for automatic eligibility for a project "A Project type is automatically eligible for Gold Standard Certification if there are Gold Standard approved Activity Requirements and/or Impact Quantification Methodologies associated with it or it's referenced in the Gold Standard Product Requirements." The Gold Standard has published Community Services Activity Requirements; therefore, the project activity falls
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		under the list of Pre-identified eligible project and is automatically eligible for Gold standard certification.
2. Location of Project	Projects may be located in any part of the world.	Location of the project is various states of India i.e. Punjab, Odisha, West Bengal, Bihar and Jharkhand.
3. Project Area, Project Boundary and Scale	The Project Area and Project Boundary shall be defined. Projects may be developed at any scale although certain rules, requirements and limitations may apply under specific Activity Requirements, Impact Quantification Methodologies and Products Requirements. In order to avoid double counting the Project shall not be included in any other voluntary or compliance standards programme unless approved by Gold Standard (for or compliance standard programme of a similar nature, the Project shall demonstrate that there is no double counting of impacts at design and performance certification (for example use of similar technology or practices through which the potential arises for double counting or misestimation of impacts amongst projects) example through dual certification). Also, if the Project Area overlaps with that of another Gold Standard or other voluntary	The boundary for the project activity in terms of a geographical area is defined as the territorial boundary of the states of Punjab, Odisha, West Bengal, Bihar and Jharkhand, India. The project falls under waste handling and disposal with annual emission reductions 46,657 tCO₂ with installed energy output of 37.15 MWth (Below the threshold of 45 MWth). Hence, the project falls under small scale projects. To avoid inclusion of any bio-digester which is a part of another registered carbon project/ programme, all the 10,702 bio-digesters under this project activity have a unique ID number to uniquely identify the bio-digesters avoiding any double counting and trace its user, later during monitoring and verification.
4. Host Country Requirements	Projects shall be in compliance with applicable Host Country's	Yes, the project is in compliance with the Host Country's legal, environmental, ecological and

	legal, environmental, ecological and social regulations.	social regulations. Project Owner has voluntarily implemented the project activity and there are no legal, environmental, ecological and social regulations /law that mandatorily enforces the implementation of the project activity or there is no other law/regulations that prevents the project owner to set up the project activity.
5. Contact Details	As part of the Project Documentation the Project Developer shall provide (i) name and (ii) contact details of all Project Participants; AND in case of an organisation (iii) the legal registration details and (iv) documentation by the governing jurisdiction that proves that the entity is in good standing (defined as being a legal or other appropriate entity registered in or allowed to operate within the required jurisdiction and with no evidence of insolvency or legal/criminal notices placed against it or any of its Directors). Gold Standard retains the right (at its own discretion) to refuse use of the Standard where reputational concerns are highlighted.	Name and Contact details of organisation is given in the Appendix 1.
6. Legal Ownership	Full and uncontested legal ownership of any Products that are generated under Gold Standard Certification, (for example carbon credits) shall be demonstrated. Where such ownership is	The project activity includes installation of household biogas plants in the rural areas of Punjab, Odisha, West Bengal, Bihar and Jharkhand. The owners of the biodigester have transferred their

	transferred from project beneficiaries this must be demonstrated transparently and with full, prior and informed consent (FPIC). Note that for certain Project types there is a requirement for full and uncontested legal land title/tenure to be demonstrated. These are contained within specific Activity or Product Requirements. All projects shall immediately report to Gold Standard any land title/tenure disputes arising.	all legal rights, interests, credits, entitlements, benefits or allowances arising from or in connection with any greenhouse gas emissions reductions arising from the operation of the digester (Emission Reduction) in the lieu of free operation and maintenance of the plant to Navrang Seva Sansthan. Same is mentioned in the Commissioning certificates co-signed by the respective plant owners. Also, Navrang Seva Sansthan has signed agreement with Greneity Infocom Service Pvt Ltd to take all necessary action required for registering the project under suitable Voluntary Standard along with complete assistance during operation and maintenance of the biogas plants included in the project activity. Also feedback has been collected during the physical local stakeholder consultation meeting as well as SFR round for the project activity.
7. Other Rights	As well as legal title and ownership, the Project Developer shall also demonstrate where required uncontested legal rights and/or permissions concerning changes in use of other resources required to service the Project (for example, access rights, water rights etc.). Any known disputes or contested rights must be declared	Not Applicable.

	immediately to Gold Standard by the Project Developer and resolved prior to further project implementation in affected areas.	
8. Official Development Assistance (ODA) Declaration	All Project Developers applying for project activities located in a country named by the OECD Development Assistance Committee's ODA recipient list and seeking Gold Standard Certification for carbon credits shall declare the Official Development Assistance (ODA) support. The Project Developer shall follow the GHG Emissions Reduction & Sequestration Product Requirements and submit the declaration at the time of Design Certification.	No ODA is involved in the Project activity. A declaration is being submitted.

Eligibility under Gold Standard Community Services Activity (CSA) Requirements

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As per section 3.1.1 of GS4GG Principles & Requirements, Eligibility criteria is defined below:

Eligibility Criteria Category	Eligibility criterion - Required condition	Justification
1. Eligible Project Types	All CSA Projects shall lead to climate change mitigation and/or adaptation by providing or improving access to services/resources at the household or community or institution level. Eligible services include electricity and energy, water and sanitation, waste management, housing, etc.	The project is already implemented since its start date. As per section 3.1.1 (a) of GG4GG Community Services Activity Requirements ver.1.2," Renewable energy types such as solar (photovoltaic and solar thermal electricity generation), tidal/wave, wind, hydropower, geothermal, waste to energy and renewable biomass that are connected to mini grid ³ or off grid solutions for targeted users and/or

		applications.” Since, project activity includes utilization of cow dung as feed to bio-digesters along with other organic waste to generate biogas Hence, it meets the criterion.
2. GENERAL ELIGIBILITY CRITERIA - Type of project	(a) Renewable energy: Renewable energy types such as solar (photovoltaic and solar thermal electricity generation), tidal/wave, wind, hydropower, geothermal, waste to energy and renewable biomass that are connected to mini grid³ or off grid solutions for targeted users and/or applications.	As per section 3.1.1 (a) of GG4GG Community Services Activity Requirements ver.1.2,” Renewable energy types such as solar (photovoltaic and solar thermal electricity generation), tidal/wave, wind, hydropower, geothermal, waste to energy and renewable biomass that are connected to mini grid³ or off grid solutions for targeted users and/or applications.” Since, project activity includes utilization of cow dung as feed to bio-digesters along with other organic waste to generate biogas Hence, it meets the criterion.
3. GENERAL ELIGIBILITY CRITERIA – Project Area, Boundary and scale	Project Area and Boundary shall be defined in line with the applicable Impact Quantification Methodologies and Product Requirements.	The boundary for the project activity in terms of a geographical area is defined as the territorial boundary of the states of Punjab, Odisha, West Bengal, Bihar and Jharkhand, India. The project falls under waste handling and disposal with emission reductions 46,657 tCO₂ with installed energy output of 37.15 MWth (Below the threshold of 45 MWth). Hence, the project falls under small scale projects.

		To avoid inclusion of any bio-digester which is a part of another registered carbon project/ programme, all the 10,702 bio-digesters under this project activity have a unique ID number to uniquely identify the bio-digesters avoiding any double counting and trace its user, later during monitoring and verification.
4. GENERAL ELIGIBILITY CRITERIA – Legal Ownership	(a) Projects involving the distribution of a large number of devices for services such as heating, cooking, lighting, electricity generation, water treatment technology such as water filter, etc. shall provide a clear description of the ownership of the Products that are generated under Gold Standard Certification all along the investment chain. In line with the FPIC requirement, the proofs that end-users are aware of and willing to give up their rights on Products shall be provided. (b) The transfer of Product ownership shall be discussed during local stakeholder consultations for projects.	The project activity includes installation of household biogas plants in the rural areas of Punjab. The owners of the biodigester have transferred their all legal rights, interests, credits, entitlements, benefits or allowances arising from or in connection with any greenhouse gas emissions reductions arising from the operation of the digester (Emission Reduction) in the lieu of free operation and maintenance of the plant to Navrang Seva Sansthan. Same is mentioned in the Commissioning certificates co-signed by the respective plant owners. Also, Navrang Seva Sansthan has signed agreement with Greneity Infocom Service Pvt Ltd to take all necessary action required for registering the project under suitable Voluntary Standard along with complete assistance during operation and maintenance of the biogas plants included in the project activity.

		Also feedback has been collected during the physical local stakeholder consultation meeting as well as SFR round for the project activity.
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A.1.2. Legal ownership of products generated by the project and legal rights to alter use of resources required to service the project

>> The project activity is the household level biogas plants and the owner of the technology is the particular household using biogas plants. The owners of a digester signed an agreement with Navrang Seva Sansthan by transferring all legal rights, interests, credits, entitlements, benefits or allowances arising from or in connection with any greenhouse gas emissions reductions arising from the operation of the digester (Emission Reduction), and agrees to take all necessary action required to ensure the transfer of those Emission Reductions to the Navrang Seva Sansthan, including executing any relevant documents.

A.2 Location of project

>>

Country: India

State: Odisha, Punjab, West Bengal, Bihar and Jharkhand State	Geo-coordinates
Odisha	20.27°N, 85.82°E
Bihar	25.4°N, 85.1°E
Jharkhand	23.35°N, 85.33°E
Bengal	22.34°N, 88.22°E
Punjab	30.47°N, 75.50°E



A.3 Technologies and/or measures

>> The existing traditional stoves used in the baseline are simple structures made from clay or having stone or metal tripods with poor combustion air supply or flue gas ventilation system i.e. without a grate or a chimney. These stoves use non-renewable biomass (firewood). The target group of the project are households with at least one head of cattle (generally cows or buffalos)

who currently use non-renewable woody biomass (firewood) for cooking purposes. The project aims to utilize the methane produced from household biogas digesters to replace the current non-renewable energy sources that is used by households.

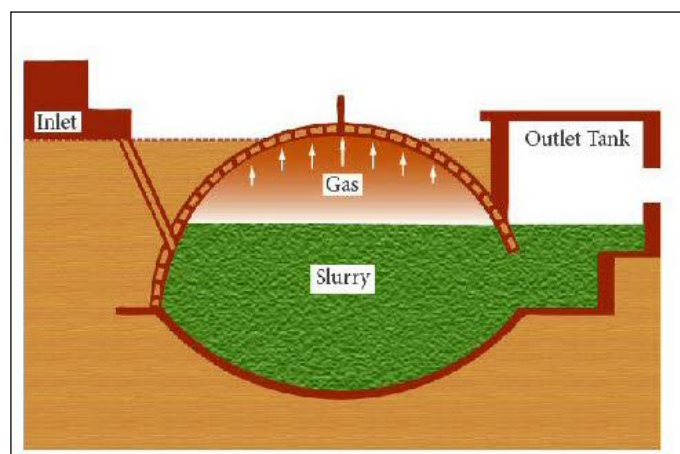
The project activity involves bundling of household bio-digesters with deenbandhu model and KVIC model technology installed in the states of Odisha, Punjab, West Bengal, Bihar and Jharkhand. The bio-digester models of various sizes (2m^3 , 3m^3 , 4m^3 and 6m^3). The major feed cattle dung is mixed with water and fed into the plant through the inlet chamber of the plant. This waste is converted into biogas with the help of a special type of anaerobic bacteria. The digested material, which comes out of the plant, is enriched manure.

Deenbandhu Model:

The major feed cow dung and other organic waste material is mixed with water and fed into the plant through the inlet chamber of the plant. This waste is converted into biogas with the help of a special type of anaerobic bacteria. The digested material, which comes out of the plant, is enriched manure.

The main feature of a Deenbandhu biogas plant is the fixed underground digester chamber, constructed with a layer of bricks and an additional layer of cement mortar forming the roof above. Connected to the underground chamber is an inlet tank (labelled on diagram as "Mixing Tank"), through which manure is fed into the plant. The manure then ferments separating the slurry from the methane gas which rises and collects at the top of the digester tank, and is released through the gas outlet pipe. The slurry passes into the outlet tank where it is ejected from the plant and can be used as fertilizer on the field

Figure1: Deenbandhu type Biogas plant



Function

A fixed-dome plant comprises of a closed, dome-shaped digester with an immovable, rigid gas-holder and a displacement pit, also named 'compensation tank'. The gas is stored in the upper part of the digester. When gas production commences, the slurry is displaced into the compensating tank. Gas pressure increases with the volume of gas stored, i.e. with the height difference between the two slurry levels. If there is little gas in the gas-holder, the gas pressure is low.

Digester

The digesters of fixed-dome plants are usually masonry structures, structures of cement and ferro-cement exist. Main parameters for the choice of material are:

- Technical suitability (stability, gas- and liquid tightness);
- availability in the region and transport costs;
- availability of local skills for working with the particular building material.

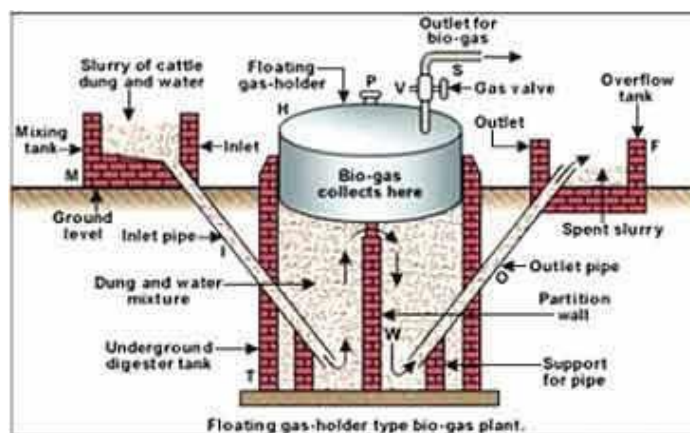
Fixed-dome Gasholder

A fixed-dome gas-holder in deenbandhu model is a hemispherical digester. The top part of a fixed-dome plant (the gas space) must be gas-tight. Concrete, masonry and cement rendering are not gas-tight. The gas space must therefore be painted with a gas-tight layer (e.g. 'Water-proofer', Latex or synthetic paints). A possibility to reduce the risk of cracking of the gas-holder consists in the construction of a weak-ring in the masonry of the digester. This "ring" is a flexible joint between the lower (water-proof) and the upper (gas-proof) part of the hemispherical structure. It prevents cracks that develop due to the hydrostatic pressure in the lower parts to move into the upper parts of the gas-holder.

KVIC floating model:

KVIC model, consists of a deep well, and a floating drum, usually made of mild steel. The system collects the gas, which is kept at a relatively constant pressure. As more gas is produced, the drum gas holder consequently rises. As the gas is consumed, the drum then falls. The biomass slurry moves through the system, as the inlet is higher than the outlet tank, creating hydrostatic pressure. Only completely digested material can flow up a partition wall, which prevents fresh material from 'short-circuiting' the system, before flowing into the outlet tank.

Figure 1: KVIC type Biogas plant



The projects operational life time is expected to be around 20 years¹.

The cattle dung otherwise would have been left to decay in open which leads to methane emissions and by utilizing the same for generation of biogas leads to capture of methane and utilize as fuel. The biogas thus generated replaces firewood that otherwise would have been used for cooking and heating purposes. The project is expected to reduce 46,657 tCO₂ per annum.

A.4 Scale of the project

>> The project activity is a small scale project activity. The project falls under waste handling and disposal with emission reductions 46,657 tCO₂ with installed energy output of 37.15 MWth (Below the threshold of 45 MWth). Hence, the project falls under small scale projects.

Requirement	Justification
Type 1: Renewable energy Projects: maximum output capacity of 15 MW(e) or 45MW(th). iii. For biomass, biofuel and biogas project activities, the maximal limit of 15 MW(e) is equivalent to a 45 MW thermal output of the equipment or the plant (e.g. boilers). For thermal applications of biomass, biofuels or biogas (e.g. cookstoves), the limit of 45 MW(th) is the installed/rated capacity of the thermal application equipment or device(s) (e.g. biogas stoves). For electrical or	The project activity falls under renewable energy project, constitute community base biogas project with installed energy output of 34.36 MWth (Below the threshold of 45 MWth). Hence, meeting the criteria of small scale projects.

¹ https://www.ctae.ac.in/images/editorFiles/file/CTAE-2017/Biogas%20FAQ_English.pdf

mechanical applications, the limit of a 15 MW installed/rated output shall be used. In the case of co-firing renewable and fossil fuels, the rated capacity of the system when using fossil fuel shall apply;	
Monitoring Methodology	The project activity used the latest small scale methodology titled AMS-I.E. 'Switch from non-renewable biomass for thermal applications by the user' version 12. Hence, meeting the criteria of <i>small scale projects</i> .

A.5 Funding sources of project

>> No public funding from parties included in Annex I to the UNFCCC, is available to the project. No Official Development Assistance involves in the project activity. The end users invest in procuring biogas technology with special assistance given by Navrang Seva Sansthan.

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

>> Methodology: AMS-I.E - Switch from non-renewable biomass for thermal applications by the user.

Version : Version 12.0

Reference: <https://cdm.unfccc.int/methodologies/DB/BLVN9ULD1FRUVS2LYWW6WPYN9W78E>

B.2. Applicability of methodology (ies)

>> AMS-I. E. "Switch from Non-Renewable Biomass for Thermal Applications by the User"

Version 12.0 is applicable to the proposed project because the project meets all applicability criteria. The applicability of the methodology is outlined as below:

Criteria AMS-I.E.	Applicability
Project participants ⁴ are able to show that non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to published literature, official reports or statistics.	Survey was conducted to check since when villagers were using firewood. It is evident from the survey that all villagers have been using firewood as a cooking fuel since 31,December 1989.
In the case that technologies using renewable biomass are used under the project activity, this methodology is applicable where all emissions related to processing of biomass are fully accounted for and biomass is sourced from biomass residues and/or a dedicated plantation of the CDM project activity, meeting the following conditions: (a) For projects that use biomass residues, prior to the implementation of the project activity, the biomass residues have not been collected and used but been left for decay and would, in the absence of the project activity, continue to be left for decay; and	Biomass is not used in the project activity. Hence, the conditions are not applicable.

(b) For projects that use biomass residues from a production process (e.g. production of sugar or wood panel boards), the implementation of the project does not result in an increase of the processing capacity of raw input (e.g. sugar, rice, logs, etc.) or in other substantial changes (e.g. product change) in this process; and (c) The biomass used by the project facility is not stored for more than one year; and (d) In the case biomass from dedicated plantations are used, the applicability conditions of TOOL16 "Project and leakage emissions from biomass" are satisfied.	
For electric cook stoves with integrated renewable energy device or with grid connected renewable energy system employing net metering, project participants shall demonstrate that, on an annual basis, at least 80% of the electricity generated is consumed by the electric cook stoves (i.e. 20% or less of electricity is consumed by other loads connected).	The project does not involve any electric cook stove and hence the condition is not applicable for the project.
For electric cook stoves, in all cases under paragraph 2(d) above where back-up diesel generators are used, this methodology is only applicable when no more than 1% of total electricity supply occurs from back up diesel generators on an annual basis.	The project does not involve any electric cook stove and hence the condition is not applicable for the project.
Under this methodology, emission reductions cannot be claimed only due to fuel-switch aspect and proposed project activities shall introduce new renewable energy based technologies, i.e. technology switch is also involved.	The project activity involves technology switch from conventional firewood based cooking system to renewable biogas based cooking system. Hence, the criteria are met.
Project participants shall describe in the PDD/PoA-DD the proposed method for distribution of project devices and how the	Each of the bio-digesters shall be allocated a unique id against each end users. End user

double counting of emission reductions has been addressed, for example, using methods such as unique identifications of product and end-user locations (e.g. programme logo), to prevent double counting of emission reductions from the project devices (e.g. between end users, distributors and producers of stoves, producers of renewable energy, producers of processed renewable biomass).	and project implementer shall have an agreement to avoid any double counting.
For project activities introducing bio ethanol cook stove, project participants shall demonstrate that the bioethanol cook stove are designed, constructed and operated to the requirements (e.g. with regard to safety) of a relevant national or local standard or comparable literature. Latest guidelines issued by a relevant national authority or an international organization may also be used.	The Project is using biogas digesters and bio- ethanol is not used and safety requirement does not arise here. Hence, the criteria is met.

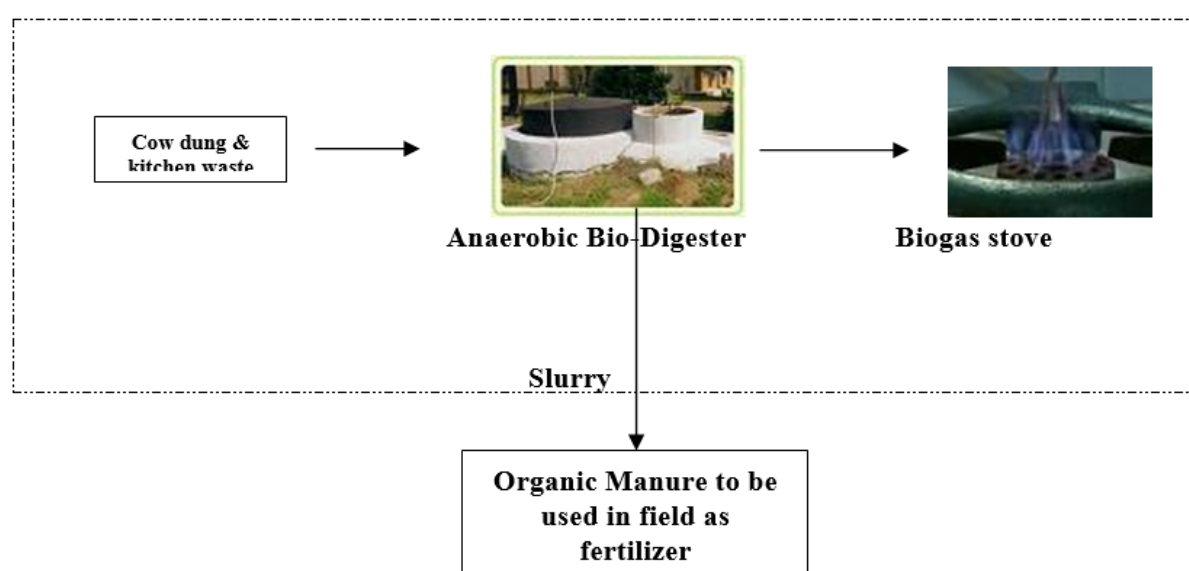
The section 3.1.1 of GS4GG Principles & Requirements states that 'A Project type is automatically eligible for Gold Standard Certification if there are Gold Standard published Activity Requirements and/or Gold Standard Approved Methodologies associated with it or as referenced in Gold Standard Product Requirements'. The GS has published Community Services Activity Requirement (Version 1.2, October 2019) therefore the proposed project falls under the automatic eligibility list of projects. The projects lead to climate change mitigation by providing access to resources (biogas) to households. The project falls under 'Renewable energy" type- Waste management and handling: Management of animal waste (cattle dung) to deliver biogas.

B.3. Project boundary

>> According to AMS-I.E, the project boundary is the physical, geographical site of the use of biomass or the renewable energy. At the unit level, the project boundary is defined by the individual sites and refers to the operation of the biogas digester units at the household level. Table below shows the emission sources that are under the control of the project participants and attributable to biogas digesters.

Source	GHGs Included?	Justification/Explanation
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Baseline	Firewood	CO ₂	Yes	Main emission source.
		CH ₄	No	Excluded for simplification.
		N ₂ O	No	Excluded for simplification.
	Emissions from fossil fuel use for cooking	CO ₂	No	Excluded for simplification.
		CH ₄	No	Excluded for simplification.
		N ₂ O	No	Excluded for simplification.
Project scenario	Firewood	CO ₂	Yes	Project emissions form continued use of fire-wood shall be accounted.
	Emission from digester and biogas cooking stove	CO ₂	No	Not required by the methodology
		CH ₄	No	Not required by the methodology
		N ₂ O	No	Not required by the methodology



B.4. Establishment and description of baseline scenario

>> As per “**AMS I.E. Switch from non-renewable biomass for thermal applications by the user, Version 12**” ‘A baseline scenario would be the use of fossil fuels for meeting similar thermal energy needs’.

The project activity involves the installation of anaerobic bio digesters for the production of biogas which replaces non-renewable biomass, used as a fuel for household cooking purposes. A baseline survey has been conducted on June 2020 to estimate the average firewood consumption prior to project activity at user point following sampling standard as per UNFCCC ‘Sampling and surveys for CDM project activities and programmes of activities’ version 08 and

guideline 'Sampling and surveys for CDM project activities and programmes of activities' version 04.

The baseline survey was performed by a third party covering various districts of Odisha, Punjab, West Bengal, Bihar and, Jharkhand. A 90% confidence interval and 10% error margin applied for sample calculation.

To understand the cooking method pattern especially with focus on targeted population in the rural areas of Odisha, Punjab, West Bengal, Bihar and, Jharkhand, to establish a baseline with respect to the fuel used for cooking; and to facilitate a clean cooking programme for the states of Odisha, Punjab, West Bengal, Bihar and, Jharkhand. As part of this objective, a primary survey of households was carried out to gauge the extent of the use of non-renewable biomass for cooking in households. The primary survey covers the targeted rural households in the state of Odisha, Punjab, West Bengal, Bihar and, Jharkhand, India.

To understand the level of detail and data points required, a conclusive research methodology, through structured face-to-face interviews, was used to arrive at statistically-valid estimates for the relevant parameters. In consonance with the objective of the study, the following methodology was adopted for carrying out the assignment.

- Preparation of detailed Questionnaire for household survey
- Mobilization of appropriate teams for village and household survey
- Orientation of the team members about the survey objectives
- Collection of primary and secondary data.
- Compilation and analysis of the collected data

Sampling Plan:

To establish a statistically representative sample size, from the total rural targeted households in Odisha, Punjab, West Bengal, Bihar and, Jharkhand, Guidelines for sampling and surveys for CDM project activities and programmes of activities (Ver04.0, CDM-EB67-A06-GUID) issued by UNFCCC, is used. As per the guidelines provided by UNFCCC, for the representative sampling methods, sample size shall be chosen for a 90/10 precision (90% confidence interval and 10% margin of error) for parameter values used to determine emission reductions.

The project activity is envisaging to implement in the states of Odisha, Punjab, West Bengal, Bihar and, Jharkhand, India. The sample population has homogeneous characteristics which are as follows:

- o Basic functioning of the bio digesters is same irrespective of their sizes.
- o All of the bio digesters are used by villagers

o Main usage of the bio digesters is to suffice domestic needs

Hence, a simple random sampling is performed. However, proportionate allocation of samples is done against the targeted population of each district. The following equation is used for arriving the sample size.

$$n \geq \frac{1.645^2 N \times p(1 - p)}{(N - 1) \times 0.1^2 \times p^2 + 1.645^2 p(1 - p)}$$

Where:

n= Sample size

N = Total number of biogas system of type i installed under the project

p = expected proportion

1.645 = represents the 90% confidence required

0.1 = represents the 10% relative precision

Using the above formula, the sample size arrived is around 263.

However, on a conservative note, the project developer, has decided to go for 476 household sampling. The total sample size decided is 476 households.

Baseline assessment:

This study examines the patterns of domestic energy consumption of targeted households in 5 states of, India over the period of past three and half decades. In India there is no legislation prohibiting the use of firewood for domestic purpose. Further, there are no legal, environmental, ecological and social regulations /law that mandatorily enforces the use of household biogas digesters.

Potential Fuels for cooking in the survey area are as follows;

Kerosene: Dependence on kerosene as a primary fuel for cooking or for lighting has decreased in India, and it is used much less for cooking than for lighting. It is evident from the survey, that none of the households utilize Kerosene as the primarily source of energy for cooking owing to is higher price and limited quantity is available is Fair price shop which is not sufficient to meet their energy demands. Prices in the open market shop is very high and hence making it out of the reach of the local population.

LPG: During this survey it was found that Fuel wood has remained the principal component of rural domestic energy. From the survey conducted it is very evident that, In India, the transition to modern fuel is slow and still a large number of households are using traditional fuels for cooking and lighting. So, it is important to understand the household level behavior towards use of different types of traditional and modern fuel. **More than 80 per cent of the households still bank upon the traditional energy sources for cooking i.e. firewood** with the slow rate of growth of LPG, the Government of India launched the Pradhan Mantri Ujjwala Yojana (PMUY) in May 2016 to facilitate underprivileged households' access to LPG for cooking (Cabinet Committee on Economic Affairs 2018). The scheme aimed to empower women and safeguard health. But still, LPG has not become the primarily source of fuel because of the following reasons:

- **High upfront costs**
- **High refill costs**
- **Extended refilling time**
- **Lack of adequate infrastructure**

During the survey it was found, there were lack of LPG distributors within a reasonable distance and therefore can be considered as major problem, as carrying cylinders over long distances which is exhausting, especially in view of the lack of roads in some rural areas are acting as the main barriers for the non-adoption of LPG.

Firewood: It was found during survey firewood provided the main fuel source for home heating, domestic hot water, and food preparation. The forest has been the dominant source for the supply of wood. However, due to increase in demand, fuel wood is also sourced from Trees outside forest, shrubs, dead logs. Being cost effective, relative affordability and easily accessible makes its primary fuel choice.

The baseline survey study concludes that, firewood is the primary source of fuel in targeted rural households in the various districts of Odisha, Punjab, West Bengal, Bihar and, Jharkhand, India.

The kitchen performance test: The kitchen performance test is performed to understand an average consumption of fuelwood usage per household. The kitchen performance test allows the testers to understand the rate of daily fuelwood consumption per household, as they are used in the normal household environment over a period of time. The KPT is a prolonged test conducted with the willing cooperation of the individual families. More than 90% of the approached families are willing to participate in the kitchen performance test. It is done by conducting daily measurement as families use the traditional stove for a period of 3 days. Results are compiled at the end of the test period. The kitchen performance test household data helps to derive the monthly and yearly consumption of biomass usage per household.

Total number of individuals in a household reflects the rationality of the quantity of biomass fuel consumed.

Average number of individuals in a household reflects the rationality of the quantity of biomass fuel consumed.

District	Average Household size
Odisha	5.59
Jharkhand	5.65
Punjab	5.58
West Bengal	5.60
Bihar	5.50

For 3 experimental days, it is evidenced that the average frequency for food preparation is 2-3 times a day. The practice of additional preparation of food items/thermal applications viz. snacks, preparation of tea, hot water etc is also been carried out.

In case of experimental households, in field, the remaining quantity of the measured fuel was weighed back on daily basis and noted down. However, for the analytical purpose, the average daily consumption for the three experimental days is taken. The survey results on household's fuel consumption are as follows.

Districts	Fuelwood usage (kg/day)	Annual Fuelwood Use (Tons)
Odisha	12.74	4.65
Jharkhand	9.59	3.50
Punjab	14.79	5.40
West Bengal	12.66	4.62
Bihar	10.54	3.85

B.5. Demonstration of additionality

B.5.1 Prior Consideration

Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable).

The project activity involves the installation of anaerobic bio digesters for the production of biogas which replaces non-renewable biomass, used as a fuel for household cooking purposes. The methodology used is AMS I.E. Switch from non-renewable biomass for thermal applications by the user, Version 12. The project falls under GG4GG Community Services Activity Requirements. As per Annex-B Positive list under 'GG4GG Community Services Activity Requirements' the project meets the criteria 3 'Project activities solely composed of isolated units where the users of the technology/measure are households or communities or institutions and where each unit results in ≤ 600 MWh of energy savings per year or ≤ 600 tonnes of emission reductions per year'.

Describe how the proposed project meets the criteria for deemed additionality.

The emission reduction for an individual unit is far below 600 tonnes of emission reduction per year. Hence, the proposed project activity falls under the 'Positive List' and hence under Principle 5 – Financial Additionality & Ongoing Financial Need, the project is considered deemed additional.

>> As per GS4GG rule for retroactive projects, project documents need to submit to GS within one year of the project start date to meet prior consideration. Only biogas plants commissioned from one year prior to GS submission shall be considered. The project activity is retroactive and the chronology of the events is mentioned in the table below;

Start date of the project	15/11/2020
Final Commissioning date	22/08/2021
Date of first submission to GS	11/11/2021
Date of initiation of SFR	18/01/2022
End date of SFR	19/03/2022
Date of Listing	27/01/2022
VVB appointment date	07/12/2021
Design Review	29/03/2022

B.5.2 Ongoing Financial Need

>> As per the clause 3.5.2.2 of GS4GG 'principle and requirements', the ongoing financial needs will be demonstrated during the Design Certification Renewal stage.

B.6. Sustainable Development Goals (SDG) outcomes

Relevant Target/Indicator for each of the three SDGs

Sustainable Development Goals Targeted	Most relevant SDG Target	SDG Impact
		Indicator (Proposed or SDG Indicator)
13 Climate Action (mandatory)	SDG target (13.2)- Integrate climate change measures into national policies, strategies and planning	Quantity of emission reductions achieved from the project activity.
8 Decent Work and Economic Growth	SDG target 8.5.2- Quality of Employment; and Quantitative employment and income generation	Number of Training Programmes provided to the employees and the Maintenance & Service staff. Number of Employees for Maintenance & Service.
7 Affordable and clean energy	7.1- By 2030, ensure universal access to affordable, reliable and modern energy services	7.1.2-Proportion of population with primary reliance on clean fuels and technology
3 Good Health and Well Being	3.9 By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination.	3.9.1Mortality rate attributed to household and ambient air pollution

B.6.1 Explanation of methodological choices/approaches for estimating the SDG Impact

>>

As per "AMS I.E- Switch from non-renewable biomass for thermal applications by the user, Version 12, the baseline emissions (BE_y) are calculated as:

$$BE_y = B_y \times f_{NRB,y} \times NCV_{biomass} \times EF_{projected_fossil_fuel}$$

Where,

BE_y = Baseline emissions during the year y in t CO₂e

B_y = Quantity of woody biomass that is substituted or displaced in tonnes

f_{NRB} = Fraction of woody biomass used in the absence of the project activity in year y that can be established as non-renewable biomass (fNRB)

$NCV_{biomass}$ = Net calorific value of the non-renewable woody biomass that is substituted (IPCC default for wood fuel, 0.0156 TJ/tonne)

$EF_{projected_fossil\ fuel}$ = Emission factor for the substitution of non-renewable woody biomass by similar consumers. Use a value of 64.4 tCO₂/TJ

B_y is determined by using option (a) paragraph 27 of the methodology as follows:

"Calculated as the product of the number of households multiplied by the estimate of average annual consumption of woody biomass per household that is displaced by the project activity (tonnes/ household/year)";

$$B_y = N_{HH} \times (BC_{BL,HH,y} - BC_{PJ,HH,y})$$

Where,

N_{HH} = Number of households in the project activity, number

$BC_{BL,y}$ = Average annual consumption of woody biomass per household before the start of the project activity, tonnes/household/year

$BC_{PJ,HH,y}$ = If it is found that pre-project devices were not completely displaced but continue to be used to some extent, average annual consumption of woody biomass per household in the pre-project devices during the project activity, tonnes/household/year

$BC_{BL,HH,y}$ for the project have been considered based on previous survey and publicly available reports as discussed in above section.

Fraction of woody biomass used in the absence of the project activity in year y that can be established as nonrenewable biomass ($f_{NRB,y}$) is determined as per methodological tool 'Calculation of the fraction of nonrenewable biomass' version 03 as follows:

The fraction of woody biomass that can be established as non-renewable, is:

$$f_{NRB} = \frac{NRB}{NRB + RB}$$

f_{NRB} = Fraction of non-renewable biomass (fraction or %)

NRB = Quantity of non-renewable biomass (t/yr)

RB = Quantity of renewable biomass (t/yr)

Estimation of consumption of woody biomass (H/Bold,total) is done following paragraph 11 (a) of the tool 'Official Statistics'.

Jharkhand

As per Forest Survey of India report 2011², annual wood fuel consumption in household sector for the state of Jharkhand = 7,693,000 (Ton/year)

As per Forest Survey of India report 2011, non-domestic wood consumption (Annual fuelwood consumption in household sector and consumption of wood in House construction, Furniture and Agriculture) for the state of Jharkhand = 14,807,40 (ton/year)

As per Forest Survey of India report 2011, Total wood consumption for the state of Jharkhand = 22,500,040 (ton/year)

Bihar

As per Forest Survey of India report 2011, annual wood fuel consumption in household sector for the state of Bihar = 11,940,000 (Ton/year)

As per Forest Survey of India report 2011, non-domestic wood consumption (Annual fuelwood consumption in household sector and consumption of wood in House construction, Furniture and Agriculture) for the state of Bihar = 30,156,330 (ton/year)

As per Forest Survey of India report 2011, Total wood consumption for the state of Bihar = 42,096,330 (ton/year)

Odisha

As per Forest Survey of India report 2011, annual wood fuel consumption in household sector for the state of Odisha = 11,865,000 (Ton/year)

As per Forest Survey of India report 2011, non-domestic wood consumption (Annual fuelwood consumption in household sector and consumption of wood in House construction, Furniture and Agriculture) for the state of Odisha = 21,164,620 (ton/year)

As per Forest Survey of India report 2011, Total wood consumption for the state of Odisha = 33,029,620 (ton/year)

Punjab

As per Forest Survey of India report 2011, annual wood fuel consumption in household sector for the state of Punjab = 3,377,000(Ton/year)

As per Forest Survey of India report 2011³, non-domestic wood consumption (Annual fuelwood consumption in household sector and consumption of wood in House construction, Furniture and Agriculture) = 32,221,700(ton/year)

As per Forest Survey of India report 2011⁴, Total wood consumption = 35,598,700(ton/year)

² https://fsi.nic.in/cover_2011/chapter7.pdf

West Bengal

As per Forest Survey of India report 2011, annual wood fuel consumption in household sector for the state of West Bengal = 20,519,000 (Ton/year)

As per Forest Survey of India report 2011, non-domestic wood consumption (Annual fuelwood consumption in household sector and consumption of wood in House construction, Furniture and Agriculture) for the state of West Bengal = 46,218,980 (ton/year)

As per Forest Survey of India report 2011, Total wood consumption for the state of West Bengal = 66,737,980 (ton/year)

Procedure to estimate RB:

Renewable biomass (RB) in the country/region/area is estimated using the equation below:

$$RB = \sum (MAI_{forest,i} \times (F_{forest,i} - P_{forest})) + \sum (MAI_{other,i} \times (F_{other,i} - P_{other}))$$

Where:

$MAI_{forest,i}$ = Mean Annual Increment of woody biomass growth per hectare in subcategory i of forest areas (t/ha/yr).

$MAI_{other,i}$ = Mean Annual Increment of woody biomass growth per hectare in subcategory i of other wooded land areas (t/ha/yr). This value is not considered due to non-availability of data.

$F_{forest,i}$ = Extent of forest in sub-category i (ha). India state of forest report, 2019 has been referred for this parameter.

$F_{other,i}$ = Extent of other wooded land in sub-category i (ha). This is not used following footnote 4 of the tool.

P_{forest} = Extent of non-accessible area⁵ (e.g. protected area where extraction of wood is prohibited, geographically remote area) within forest areas (ha). This parameter is optional and not considered.

P_{other} = Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within other wooded land areas (ha). This parameter is optional and not considered.

I = Sub-category i of forest areas and other wooded land areas

State	Jharkhand ⁶	Bihar ⁷	Odisha ⁸	Punjab ⁹	West Bengal ¹⁰

⁵ CDM Methodological Tool -30: Calculation of the fraction of non-renewable biomass, version 03

⁶ <https://fsi.nic.in/isfr19/vol2/isfr-2019-vol-ii-jharkhand.pdf> (Page no. 112)

⁷ <https://fsi.nic.in/isfr19/vol2/isfr-2019-vol-ii-bihar.pdf> (Page no. 34)

⁸ <https://fsi.nic.in/isfr19/vol2/isfr-2019-vol-ii-odisha.pdf> (Page no. 202)

⁹ <https://fsi.nic.in/isfr19/vol2/isfr-2019-vol-ii-punjab.pdf> (page no. 212 of the document)

¹⁰ <https://fsi.nic.in/isfr19/vol2/isfr-2019-vol-ii-west-bengal.pdf> (Page no. 294)

¹¹ MAI _{forest,i}	0.5	0.5	0.5	0.5	0.5
F _{forest,i}	2239000	622000	5814000	256000	1173000
Renewable biomass (RB)	1119500	311000	2907000	147,500	586500

The fraction of woody biomass that can be established as non-renewable, is:

$$f_{NRB} = \frac{NRB}{NRB + RB}$$

Accordingly, the fNRB would be:

State	H	RB	NRB (H-RB)	fNRB
Jharkhand	22,500,040	1,119,500	21,380,540	95%
Bihar	42,096,330	311,000	41,785,330	99%
Orissa	33,029,620	2,907,000	30,122,620	91%
Punjab	35,470,700	128,500	35,470,700	99%
West Bengal	66,737,980	586,500	66,151,480	99%

Project Emissions (PEy):

As per applied methodology AMS-I.E, version 12, project emissions are accounted for below activities:

- CO₂ emissions from on-site consumption of fossil fuels due to the project activity
- CO₂ emissions from electricity consumption by the project activity
- Methane emission from solid waste disposal or waste water
- Project emissions related to cultivation of feedstock
- Project emissions from transportation

The project activity does not involve any of the above activity and hence, project emissions for the project activity is not applicable. However, while determining By as per equation 3 of the applied methodology, firewood consumed by pre-project devices during the project activity shall be monitored and applied ex-post. This is to be accounted.

Leakage Emissions (LEy):

Leakage emissions (related to the non-renewable woody biomass saved by the project activity) shall be assessed based on ex post surveys of users and the areas from which this woody biomass is sourced (using 90/30 precision for a selection of samples). The following potential source of leakage shall be considered: The use/diversion of non-renewable woody biomass saved under the project activity by non-project households/users that previously used renewable

¹¹ <https://moef.gov.in/wp-content/uploads/2019/06/Pacific.pdf> (page no. 16 of Asia-Pacific Forestry Sector Outlook Study II India Outlook Study 2020)

energy sources. If this leakage assessment quantifies an increase in the use of non renewable woody biomass used by the non-project households/users that is attributable to the project activity, then B_y is adjusted to account for the quantified leakage. Alternatively, B_y is multiplied by a net to gross adjustment factor of 0.95 to account for leakages, in which case surveys are not required.

PP has opted default option, and B_y shall be adjusted with adjustment factor of 0.95 to account leakage.

Emission reductions:

Emission reductions are to be estimated based on the equation below:

$$ER_y = BE_y - PE_y - LE_y$$

B.6.2 Data and parameters fixed ex ante

Data/parameter	N _{HH} (SDG 13)	
Unit	Number	
Description	Number of households in the project activity in year y	
Source of data	Project Proponent’s project database	
Value(s) applied	10,702	
	Odhisha	1408
	Jharkhand	578
	Punjab	4250
	West Bengal	1400
	Bihar	3066
Choice of data or Measurement methods and procedures	Project database as per the commissioning reports and end user agreements	
Purpose of data	Estimation of Baseline	
Additional comment	NIL	

Data/parameter	$BC_{BL,HH,y}$ (SDG 13)
----------------	-------------------------

Unit	tonnes/household/year										
Description	Average annual consumption of woody biomass per household before the start of the project activity										
Source of data	Baseline survey										
Value(s) applied	<table> <tr> <td>Odhisia</td><td>4.65</td></tr> <tr> <td>Jharkhand</td><td>3.50</td></tr> <tr> <td>Punjab</td><td>5.40</td></tr> <tr> <td>West Bengal</td><td>4.62</td></tr> <tr> <td>Bihar</td><td>3.85</td></tr> </table>	Odhisia	4.65	Jharkhand	3.50	Punjab	5.40	West Bengal	4.62	Bihar	3.85
Odhisia	4.65										
Jharkhand	3.50										
Punjab	5.40										
West Bengal	4.62										
Bihar	3.85										
Choice of data or Measurement methods and procedures	To know the baseline fuel/technologies which is the firewood and its consumption pattern, a survey was conducted in the states of Punjab, Orissa, Jharkhand, West Bengal and Bihar. The survey is conducted in line with the methodology and following UNFCCC sampling standard 'Sampling and surveys for CDM project activities and programmes of activities' version 08 and guideline 'Sampling and surveys for CDM project activities and programmes of activities' version 04. A 90% confidence interval and 10% error margin applied for sample calculation.										
Purpose of data	Estimation of Baseline										
Additional comment	NIL										

Data/parameter	$f_{NRB,y}$ (SDG 13)						
Unit	Percentage						
Description	Fraction of woody biomass saved by the project activity during year y that can be established as non-renewable biomass						
Source of data	Calculated as per the requirements of methodological tool - Calculation of the fraction of non-renewable biomass, version 03.						
Value(s) applied	<table> <tr> <td>Orissa</td><td>91%</td></tr> <tr> <td>Jharkhand</td><td>95%</td></tr> <tr> <td>Punjab</td><td>99%</td></tr> </table>	Orissa	91%	Jharkhand	95%	Punjab	99%
Orissa	91%						
Jharkhand	95%						
Punjab	99%						

	West Bengal	99%
	Bihar	99%
Choice of data or Measurement methods and procedures	Calculation of the fraction of non-renewable biomass, version 03 (Methodological tool)	
Purpose of data	Estimation of Baseline	
Additional comment	NIL	

Data/parameter	NCV _{biomass} (SDG 13)	
Unit	TJ/tonne	
Description	Net calorific value of the non-renewable woody biomass that is substituted	
Source of data	Default value from the methodology, AMS-I.E - Switch from non-renewable biomass for thermal applications by the user, Version 12.0.	
Value(s) applied	0.0156 TJ/tonne	
Choice of data or Measurement methods and procedures	Default value from the methodology, AMS-I.E - Switch from non-renewable biomass for thermal applications by the user, Version 12.0.	
Purpose of data	Estimation of Baseline	
Additional comment	NIL	

Data/parameter	EF _{projected_fossilfuel} (SDG 13)	
Unit	tCO ₂ /TJ	
Description	Emission factor for the substitution of non-renewable woody biomass	
Source of data	Default value from the methodology, AMS-I.E - Switch from non-renewable biomass for thermal applications by the user, Version 12.0.	

Value(s) applied	64.4 tCO ₂ /TJ
Choice of data or Measurement methods and procedures	Default value from the methodology, AMS-I.E - Switch from non-renewable biomass for thermal applications by the user, Version 12.0.
Purpose of data	Estimation of Baseline
Additional comment	NIL

B.6.3 Ex ante estimation of SDG Impact

>>

Baseline Emissions:

$$BE_y = B_y \times f_{NRB,y} \times NCV_{biomass} \times EF_{projected_fossil_fuel}$$

Where,

$$B_y = NHH \times (BC_{BL,HH,y} - BC_{PJ,HH,y})$$

States	Total population (N _{HH})	Average firewood consumption (ton/year) BC _{BL,HH,y}	By (Ton/year)
Orissa	1408	4.65	6547.2
Jharkhand	578	3.50	2023
Punjab	4250	5.40	22950
West Bengal	1400	4.62	6468
Bihar	3066	3.85	11804.1

BC_{Pj,HH,y} (if it is found that pre-project devices were not completely displaced but continue to be used to some extent, average annual consumption of woody biomass per household in the pre-project devices during the project activity, tonnes/household/year) is considered zero during design certification stage. The same shall be monitored ex-post certification.

Fraction of woody biomass that can be established as non-renewable for the states are as follows (*f_{NRB,y}*)

Orissa	91%
Jharkhand	95%
Punjab	99%
West Bengal	99%

Bihar	99%
-------	-----

BEy would be:

$$BEy = By \times fNRB_y \times NCV_{biomass} \times EF_{projected_fossil_fuel}$$

States	By (Ton/year)	NCV	EF	BEy (tCO ₂ e)
Orissa	6547.2	0.0156	64.4	5998
Jharkhand	2023	0.0156	64.4	1931
Punjab	22950	0.0156	64.4	22973
West Bengal	6468	0.0156	64.4	6440
Bihar	11804.1	0.0156	64.4	11771
Total				49,113

Project Emissions (PEy): As explained in above, project emission is zero.

Leakage Emissions: By is multiplied by a net to gross adjustment factor of 0.95 to account for leakages. Therefore, Leakage would be 2,455.65 tCO₂e

$$\begin{aligned} \text{Emission reductions} &= BEy - PEy - LEy \\ &= 49,113 - 0 - 2,455.65 \\ &= 46,657 \text{ tCO}_2\text{e} \end{aligned}$$

SDG 7:	
Baseline estimation: All users used fire-wood based cooking system which is not clean energy.	10,702 users are accessed to clean energy sources. This shall be monitored. It is assumed that with 100% operational status all users are accessed to clean energy source.
SDG 3:	
Prone to illness due to smoke released from burning of fire-wood.	Reduction in smoke related illness due to use of biogas instead of firewood. Since, biogas is smokeless; 10,702 users are expected to get rid of kitchen smoke related illness.
SDG 8:	
Less local employment opportunities for youth.	15 permanent employments, excluding more than 20 contractual employments. Creates more local employment opportunities.
Unskilled local villagers; and less employability.	2 trainings per year. More trained/skilled youth.

B.6.4 Summary of ex ante estimates of each SDG Impact

Year	Baseline estimate	Project estimate	Net benefit (after Leakage adjustment)
Year 1	49,113	0	46,657
Year 2	49,113	0	46,657
Year 2	49,113	0	46,657
Year 4	49,113	0	46,657
Year 5	49,113	0	46,657
Total	245,565	12,278.25 0	233,286
Total number of crediting years	5		
Annual average over the crediting period	49,113	0	46,657

SDG 8

Year	Baseline estimate (number of people employed by the project)	Project estimate (number of people employed by the project)	Net benefit (number of people employed by the project)
Year 1 to Year 5	0	At least 15 people	At least 15 people
Total	0	At least 15 people	At least 15 people
Total number of the crediting years	5 years		

Year	Baseline estimate (number of trainings conducted)	Project estimate (number of trainings conducted)	Net benefit (number of trainings conducted)
Year 1	0	2	2
Year 2	0	2	2
Year 3	0	2	2

Year 4	0	2	2
Year 5	0	2	2
Total	0	10	10
Total number of the crediting years	5 years		

SDG 3:

Year	Baseline estimate (number of people using biogas system)	Project estimate (number of people using biogas system)	Net benefit (number of people using biogas system)
Year 1 to Year 5	0	10,702	10,702
Total	0	10,702	10,702
Total number of the crediting years	5 years		

SDG 7:

Year	Baseline estimate (number of families displaced firewood)	Project estimate (number of families displaced firewood)	Net benefit (number of families displaced firewood)
Year 1 to Year 5	0	10,702	10,702
Total	0	10,702	10,702
Total number of the crediting years	5 years		

B.7. Monitoring plan

B.7.1 Data and parameters to be monitored

Greneity infocom Service Pvt Ltd (hereafter Greneity) facilitates the End User families to set up village level institutions to take care of repairs, maintenance and the social controls/peer support needed to cope with various exigencies that crops up.

Biogas Project Management Unit

A dedicated team is set up within Greneity for management and monitoring of the Biogas Project. This Project Management & Monitoring Unit consist of the following staff:

Biogas Field Workers/RETs:

Biogas field workers oversee the operation & maintenance of biogas units. Each field worker is in-charge of approximately 500 units. They report to the Project Manager and their tasks are to:

- Inspect the bio-digesters in periodic intervals to ensure proper functioning of all the plants
- Identify local suppliers of material for village clusters
- Record the working of bio-digesters in specially designed formats and handover the data to manager on a regular basis
- Assist in selection of Volunteers, train and support them
- Contract Masons and attend to major repairs that cannot be handled by the village level institutions

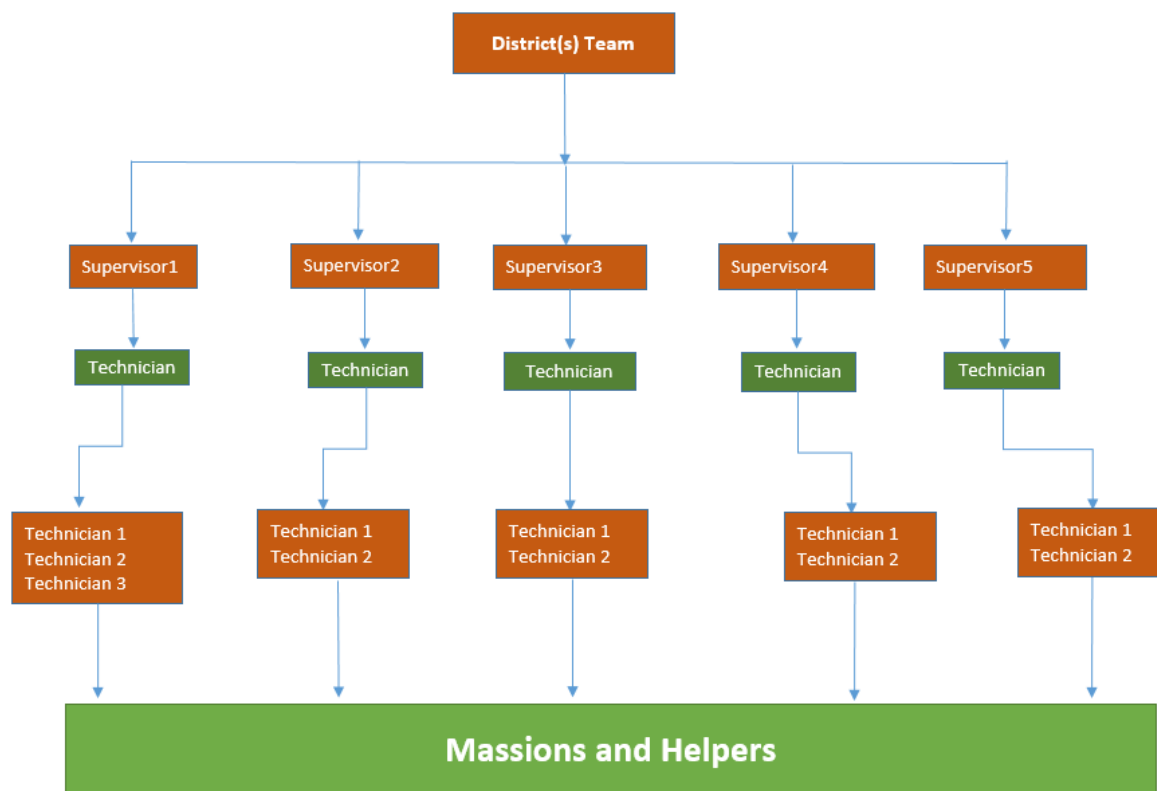
Management system at village level:

Village level Volunteers maintain the Monitoring log Books where the reasons for non-usage are recorded. Every participating village have a Volunteer to monitor usage of biogas units. Volunteer(s)/mason(s) will be responsible for monitoring of usage and is the first to identify dysfunctional units. The Volunteer either prompts the End User to set right a problem or bring it to the notice of the Biogas Technician.

Each Biogas Unit is marked with a unique Identification Number and date of construction.

A Monitoring log Book is maintained for the project activity. If any Biogas Unit is faulty or not functional, the problem report is automatically passed on to Volunteer(s)/mason(s). There is a continuous database maintained of all the Biogas Units not operational on a regular basis where number of non-operational days, and the reasons of non-operation are recorded. Mostly, volunteers are able to handle the problem arises, in case of major issues, Biogas technicians are informed to handle the situation.

Information flow diagram is as follows;



Data / Parameter	$N_{i,y}$ (SDG 13)
Unit	Number
Description	Number of project devices of type i operational in year y
Source of data	Monitoring
Value(s) applied	To be monitored
Measurement methods and procedures	Determined for all end-users or based on a representative sample (e.g.using survey methods). The "Standard: Sampling and surveys for CDM project activities and programme of activities" shall be used for determining the sample size to achieve 90/10 confidence/precision levels.
Monitoring frequency	At least once in every two years.
QA/QC procedures	Third party survey to be conducted following standard sampling approach
Purpose of data	Estimation of Baseline Emissions.
Additional comment	

Data / Parameter	$BC_{PJ,HH,y}$ (SDG 13)
Unit	tonnes/household/year
Description	Average annual consumption of woody biomass per household in the pre-project devices during the project activity, if it is found that pre-project devices were not completely displaced but continue to be used to some extent.
Source of data	Survey
Value(s) applied	To be monitored
Measurement methods and procedures	Monitoring shall consist of estimation of all project devices or a representative sample thereof, at least once every two years.
Monitoring frequency	At least once in every two years.
QA/QC procedures	Third party survey to be conducted following standard sampling approach.
Purpose of data	Estimation of Baseline Emissions.
Additional comment	Nil

Data / Parameter	N_{HH} (SDG 13)
Unit	Number
Description	Number of households (biogas system) in the project activity in operational per year
Source of data	Survey
Value(s) applied	To be monitored
Measurement methods and procedures	Monitoring consist of checking of representative sample, to ensure that bio-digesters operating. Calculated as $NHH \cdot P_y$, where NHH is the number of biodigesters installed in the project and P_y is Proportion of Bio-digesters operational estimated based on the sample survey.
Monitoring frequency	At least once in every two years.

QA/QC procedures	Third party survey to be conducted following standard sampling approach
Purpose of data	Estimation of Baseline Emissions.
Additional comment	

Data / Parameter	(SDG 8) Unemployment rate, by sex, age and persons with disabilities
Unit	Numbers
Description	Provided training for improving employability. Thereby improving Quality of Employment.
Source of data	Records of the training programmes, awareness generation activities, photographs, interviews etc.
Value(s) applied	2
Measurement methods and procedures	-
Monitoring frequency	Annual
QA/QC procedures	Continuation of regular trainings/workshops.
Purpose of data	Assessment of contribution towards 'Safeguarding Principles' and to monitor the contribution to SDG 8
Additional comment	

Data / Parameter	(SDG 8) Quantitative employment and income generation (8.5.2)
Unit	Numbers
Description	Number of people employed for the maintenance and servicing of biodigesters.
Source of data	Employment records, employment contracts maintained by Project Developer
Value(s) applied	
Measurement methods and procedures	Manually by Project Developer

Monitoring frequency	Annual
QA/QC procedures	Income generation to be enhanced by creating job opportunities through training and capacity building.
Purpose of data	To monitor the contribution to SDG 8
Additional comment	

Data / Parameter	(SDG 7) Access to affordable and clean energy services (7.1.2)
Unit	Numbers
Description	Number of biogas system operational under the project activity
Source of data	Biogas User Survey
Value(s) applied	To be monitored
Measurement methods and procedures	Sample survey to confirm if project biogas systems are operational. Operational status will confirm that the users are accessed to affordable and clean energy
Monitoring frequency	At least once in two years
QA/QC procedures	Required sample size shall be determined following UNFCCC sampling standard
Purpose of data	To monitor the contribution to SDG 7
Additional comment	Nil

Data / Parameter	(SDG 3)Improvement in health and decrease in illness (3.9.1)
Unit	Numbers
Description	Decrease in illness
Source of data	Biogas User Survey
Value(s) applied	To be monitored
Measurement methods and procedures	Improvement in health and decrease in illness will be assessed through interview with end users due to project implementation. Users opinion on indoor air quality due

	to biogas usage shall be collected during monitoring survey.
Monitoring frequency	At least once in two years
QA/QC procedures	Required sample size shall be determined following UNFCCC sampling standard
Purpose of data	To monitor the contribution to SDG 3
Additional comment	Nil

B.7.2 Sampling plan

>>

A third party sample survey has performed. The survey is conducted in line with the methodology and following UNFCCC sampling standard 'Sampling and surveys for CDM project activities and programmes of activities' version 08 and guideline 'Sampling and surveys for CDM project activities and programmes of activities' version 04. A 90% confidence interval and 10% error margin applied for sample calculation.

B.7.3 Other elements of monitoring plan

>>N/A

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1 15/11/2020

>>

C.1.2 Expected operational lifetime of project

>> 20Years

C.2. Crediting period of project

C.2.1 Start date of crediting period

>>15/11/2020 or two years prior to the date of project design certification whichever is later.

C.2.2 Total length of crediting period

>>The project falls under GS community service activity and hence eligible for total 15 years issuance with two times certification renewal every 5 years. So the length of crediting period will be 15 years.

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1 Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in [Appendix 1](#), ongoing monitoring is summarized below.

Principles	Mitigation Measures added to the Monitoring Plan
Not applicable	No safeguarding principle is adversely impacted by the project activity.

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	The project is aimed to replace conventional wood fired mud stoves technology with biogas primarily for cooking. This will result in reducing use of firewood consumption or replacement of complete firewood usage at user place. Primarily in rural areas of India, cooking activity at household level is managed by women. Therefore, women are more exposed to the indoor air pollution and the associated hazard. Women in rural areas in most cases are also responsible for taking care of their children specially infants who need mother's support most of the time are bound to accompany their mother in kitchen. This situation leads to enhanced exposure of the women and children to kitchen smoke and associated health consequences. Since the project aims to replace the polluting traditional cooking stoves with biogas cooking system, the primary beneficiary would be the women and children. Furthermore,
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	the project is focused to the economically disadvantaged group of people, which also justifies the dimension of social inclusion in the project design.
Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	Ministry of women & child development, Govt. of India has taken various measures for gender equality/socio-economic development/empowerment of women. Out of these, the project positively contributes towards the national mission for empowerment of women through improvement of health and attaining vision for empowerment of women under national policy for women 2016 (Women participation will be ensured in the efficient use and spreading the use of solar energy, biogas, smokeless chulas and other technological applications to have positive influence on their life styles and a long term impact on meeting sustainable development goals).
Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	N/A
Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	N/A

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

The below is a summary of the 2 step GS4GG Consultation for monitoring purposes. Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

E.1 Summary of stakeholder mitigation measures

Greneity and its partner NGOs invited all the relevant stake holders like Gold standard supporting NGOs, DNA, local Government officials, elected representatives, and the villagers for this meeting. The local stake holder consultation meeting was successfully held at different places in Punjab, Orissa, Bihar, West Bengal, and Jharkhand which are detailed below:

State	Date of Meeting	Place
Bihar	19 th Jan 2022, 11 AM onwards	Near Panchayat Office Village Navada Block Kaler Arwal District Bihar, India.
Orissa	20 th Jan 2022, 10 AM onwards	Kalyan Mandap Pikol Panchayat Cuttack, India
Jharkhand	23 rd Jan 2022, 10 AM onwards	Pandekesho Village Block Sariyahat Dumka Jharkhand, India
West Bengal	28 th Jan 2022, 10 AM onwards	Village Pathordihi Siuri Post, Birbhum West Bengal, India
Punjab (Bathinda)	16 th Feb 2022, 10 AM onwards	Near Panchayat Office Village Sidhana, Phool Tehsil District Bathinda, Punjab
Punjab (Mansa)	17 th Feb 2022, 10 AM onwards	Panchayat Office Village Burj, Dhalwan District Mansa, Punjab

The project idea and its components were thoroughly discussed, feedback from the participants has been documented and sustainable development aspects were discussed during the proceedings. Participants got clarification on many of the project aspects.

Village Sarpanch and Pradhan explained about the biogas program of the Government and its limitations and implementation style. They stated that the present project is much useful to the farming community in general and women in particular. Further, the positive impact on the women health by using the clean fuel technology is also explained.

RET's introduced the present project to the participants and clarified many of their queries. The RET, also explained the transfer of ownership of Carbon credits, to provide free operation and maintenance of the biogas plant. Staff members of the Navrang Sewa Sansthan recorded attendees' details in a special register. Further, RET's explained how this Project activity is supporting the installation of biogas plants to rural households and provide participating

households with clean, convenient, and efficient means for cooking and heating purposes. They also provided insights on how the monitoring of installed domestic biogas units could lead to positive sustainable development goals.

They had distributed feedback forms among the participants and collected once they were duly filled. The minutes of the meeting were recorded, and meeting photos were taken to document the event.

The following were the sample questions raised in the meeting:

Whom to contact in case of emergency breakdown?

Minimum age required to be a volunteer?

The following were the sample comments from the feedback forms:

1. The meeting is well organized and very informative.
2. The project is a needed and useful for both farmers and women.
3. There is nothing to dislike in the project.

The participants overwhelmingly supported the project, and No negative comments were raised. Stakeholder feedback round was initiated on 18/01/2022 till 19/03/2022. However, no comments were received during this period.

E.2 Final continuous input / grievance mechanism

	Method Chosen (include all known details e.g. location of book, phone, number, identity of mediator)	Justification of Choice (best practice)
Continuous Input / Grievance Expression Process Book (mandatory)	The logbook has been maintained in the panchayat office.	The panchayat office is the most accessible place for all the villagers
GS Contact (mandatory)	help@goldstandard.org	
Telephone access (optional)	+91 9886252062	As the project is spread across a huge area hence telephone access has also been provisioned.
Internet/email access (optional)	shivani.garg@greeneity.com	The stakeholders with internet access has an option of contacting Greeneity through the email id provided.
Nominated Independent Mediator (optional)	N/A	N/A
Other		

APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into [SECTION D](#) above. Please refer to the instructions in the [Guide to Completing](#) this Form.

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights 2. The Project shall not discriminate with regards to participation and inclusion	No	1. The Project is not in conflict with the economic livelihood or other issue of the local community. Thus, the Project does not cause any human rights abuse and respects internationally proclaimed human rights issue. 2. Project activities are not expected to cause any human rights abuse. As a member of United	Not required

		Nations ¹² and part of UN Agreement on Human Rights ¹³ , it is ensured by law in India that no action can be taken against human rights.	
Principle 2. Gender Equality			
1. The Project shall not directly or indirectly lead to/contribute to adverse impacts on gender equality and/or the situation of women 2. Projects shall apply the principles of nondiscrimination, equal treatment, and equal pay for equal work 3. The Project shall refer to the country's national	No	1. The project is implemented at households. It does not involve any women workforce which may lead to sexual harassment. The project does not restrict of women's rights or access to resources. The project recognizes women's	Not required

¹² <https://labour.gov.in/lcandilasdivision/india-ilo>

¹³ <https://www.ilo.org/newdelhi/lang--en/index.htm>

gender strategy or equivalent national commitment to aid in assessing gender risks 4. (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)		ownership rights regardless of marital status. 2. Yes. The project involves construction of biogas digesters at households. Trained labors are used for the same. Local people are engaged for the same. No discrimination either in gender or any other form is followed to engage local people. 3. India ratified the International Convention on the Elimination of All Forms of Racial Discrimination¹⁴ on 03/12/1968 with certain reservation. The project activity is in line with strategy of elimination of discrimination. 4. Summary of opinions and recommendations of an Expert	
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¹⁴ https://nhrc.nic.in/documents/india_ratification_status.pdf

		Stakeholder is required in case of Gender responsive project activity which is not the case for the project activity.	
Principle 3. Community Health, Safety and Working Conditions			
1. The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community	No	The project leads to safe working condition and improvement in health as it will replace firewood as fuel with biogas which is clean and safe. All the participants/staffs are well trained and are well aware of the in operational and maintenance issues as well as health & safety related topics.	Not required
Principle 4.1 Sites of Cultural and Historical Heritage			
Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture?	No	The project area covers households which does not have any structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture. No cultural heritage/ indigenous people are replaced by the project. Hence, not applicable.	Not required
>>			

Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No	The project activity takes place within individual households. Therefore, no relocation of people takes place.	Not required
>>			
Principle 4.3 Land Tenure and Other Rights			
a. Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership?	No	There are no uncertainties regarding land tenure, access rights, usage rights or land ownership. The biogas digesters are developed for the users and will belongs to them. They will have the all the rights to use and maintain the same.	Not required
b. For Projects involving land use tenure, are there any uncertainties with regards to land tenure, access rights, usage rights or land ownership?			
>>			
Principle 4.4 - Indigenous people			
Are indigenous peoples present in or within the area of influence of the Project and/or is the Project located on	No	No. The project involves household biogas digesters in the state of Punjab, India. Therefore, it does not involve	Not required

land/territory claimed by indigenous peoples?		any negative influence towards indigenous people	
>>			
Principle 5. Corruption			
1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects	No	Project activity is an installation of biogas digesters at households in the state of Punjab, India and does not contribute to or reinforce corruption of any kind. Indulgence in corruption is an illegal activity in the host country and the local labor compliance takes into account of the same. PP does not involve and is not complicit in any kind of corruption. India has ratified UN convention against Corruption in 2011 ¹⁵ .	Not required
Principle 6.1 Labour Rights			

¹⁵ https://treaties.un.org/Pages/ViewDetails.aspx?src=IND&mtdsg_no=XVIII-14&chapter=18&clang=_en#EndDec

1. The Project Developer shall ensure that all employment is in compliance with national labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions 2. Workers shall be able to establish and join labour organisations 3. Working agreements with all individual workers shall be documented and implemented and include: a) Working hours (must not exceed 48 hours per week on a regular basis), AND	No	1. The project proponent is committed to the employee's workplace health & safety during all phases of the project. All employees will attend health & safety trainings. This is issued in the Labour code on Occupational Safety, Health and Working Conditions and UN Agreement on Human Rights¹⁶. 2. The project respects fundamental right of employee. There is law in India since 1926 by The Trade Unions Act, 1926¹⁷ which protects rights of industrial trade unions and their members. PP and appointed contractors will not involve in any form of	Not required
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¹⁶ <https://www.ohchr.org/en/countries/india>

¹⁷ <http://ncw.nic.in/acts/TheTradeUnionsAct1926.pdf>

b) Duties and tasks, AND c) Remuneration (must include provision for payment of overtime), AND d) Modalities on health insurance, AND e) Modalities on termination of the contract with provision for voluntary resignation by employee, AND f) Provision for annual leave of not less than 10 days per year, not including sick and casual leave. 4. No child labour is allowed (Exceptions for children working on their families' property		forced or compulsory labor. India has ratified ILO "C029 – Forced Labor Convention"¹⁸. 3. PP and their subcontractors complying with all relevant national laws regarding child labor. PP will not employ children in any shape or form for their works. India has ratified ILO "C138 – Minimum Age Conventions" and "C182 – Worst Forms of Child Labor Convention"¹⁹. 4. PP and their subcontractors complying with all relevant national laws regarding child labor. PP will not employ children in any shape or form for their works. India has ratified ILO "C138 – Minimum Age	
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¹⁸ https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:11200:0::NO::P11200_COUNTRY_ID:102691

¹⁹ https://www.ilo.org/dyn/normlex/en/f?p=1000:11200:0::NO:11200:P11200_COUNTRY_ID:102691

requires an Expert Stakeholder opinion) 5. The Project Developer shall ensure the use of appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures		Conventions” and “C182 – Worst Forms of Child Labor Convention”²⁰. 5. The project owner is committed to the safe and healthy working conditions all phases of the project. All employees will attend trainings health & safety. This issue is protected by Labor code²¹ and UN Agreement on Human Rights²².	
Principle 6.2 Negative Economic Consequences			
1. Does the project cause negative economic consequences during and after project implementation?	No	The project has lifetime of 20 years and expected to run beyond the project certification period. With carbon finance it is further strengthen to operate 100% project systems with	Not required
>>			

²⁰ https://www.ilo.org/dyn/normlex/en/f?p=1000:11200:0::NO:11200:P11200_COUNTRY_ID:102691

²¹ <https://www.ohchr.org/en/countries/india>

²² <https://www.ohchr.org/en/countries/india>

		continuous maintenance. Thus, there is no negative economic consequences during and after project implementation.	
Principle 7.1 Emissions			
Will the Project increase greenhouse gas emissions over the Baseline Scenario?	No	No. The project will replace firewood use with biogas. Hence, it will reduce greenhouse gas emissions over the Baseline Scenario.	Not required
>>			
Principle 7.2 Energy Supply			
Will the Project use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?	No	No. The project uses inhouse cattle dung only. Cowdung and other household waste shall be used as a feed source for the digesters. There is no use of Wood or Biomass in the project activity.	Not required
>>			
Principle 8.1 Impact on Natural Water Patterns/Flows			
Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow	No	The project activity is installation of biogas digesters for energy generation to meet domestic requirements. Individual household based	Not required

variability, flooding potential, lack of aquatic connectivity or water scarcity?		biogas digesters are planned to be installed which are small in size and will not have any impact on the natural or pre-existing patter of water courses.	
>>			
Principle 8.2 Erosion and/or Water Body Instability			
a. Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion?	No	No. The risk of erosion is unlikely by the project.	Not required
b. Is the Project’s area of influence susceptible to excessive erosion and/or water body instability?			
>>			
Principle 9.1 Landscape Modification and Soil			
Does the Project involve the use of land and soil for production of crops or other products?	No	The project will not involve use of land and soil for production of crops or other products.	Not required
>>			
Principle 9.2 Vulnerability to Natural Disaster			

Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions?	No	No. The project happens at individual households. There is no activity which can affect adversely the natural system to cause earthquake, landslides, erosion, flooding, draught or other extreme climatic conditions.	Not required
>>			
Principle 9.3 Genetic Resources			
Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include GMOs in their processes and production)?	No	Not applicable. The project does not involve any crop production or cultivation.	Not required
>>			
Principle 9.4 Release of pollutants			
Could the Project potentially result in the release of pollutants to the environment?	No	The project activity is installation of biogas digesters for energy generation to meet domestic requirements. The project would lead to improving of indoor air	Not required
>>			

		pollution that occurs due to use of wood based chula's or cookstoves in the baseline. However, Bio Slurry is managed in appropriate manner as a manure. Beneficiaries using bio slurry on their land to improve soils fertility.	
Principle 9.5 Hazardous and Non-hazardous Waste			
Will the Project involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials?	No	The project activity is installation of biogas digesters for energy generation to meet domestic requirements. There are no hazardous waste generated due to the project. The slurry generated as part of the biodigester shall be used as fertilizer.	Not required
>>			
Principle 9.6 Pesticides & Fertilisers			
Will the Project involve the application of pesticides and/or fertilisers?	No	The project activity is installation of biogas digesters for energy generation to meet domestic requirements. There are no hazardous waste generated due to the project. The slurry generated as part of	Not required
>>			

		the biodigester shall be used as fertilizer.	
Principle 9.7 Harvesting of Forests			
Will the Project involve the harvesting of forests?	No	Not applicable. The project happens at individual households.	Not required
>>			
Principle 9.8 Food			
Does the Project modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives?	No	Not applicable. The project activity is biogas digester installation happens at individual households.	Not required
>>			
Principle 9.9 Animal husbandry			
Will the Project involve animal husbandry?	No	Not applicable. The project activity is biogas digester installation happens at individual households.	Not required
>>			
Principle 9.10 High Conservation Value Areas and Critical Habitats			
Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats,	No	Not applicable. The project activity is biogas digester installation happens at individual households.	Not required

landscapes, key biodiversity areas or sites identified?			
>>			
Principle 9.11 Endangered Species			
a. Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)?	No	Not applicable. The project activity is biogas digester installation happens at individual households.	Not required
b. Does the Project potentially impact other areas where endangered species may be present through transboundary affects?			
>>			

APPENDIX 2- CONTACT INFORMATION OF PROJECT PARTICIPANTS

Organization name	Greneity Infocom Service Pvt Ltd
Registration number with relevant authority	CIN-U74900DL2013PTC253219 under Companies Act,1956
Street/P.O. Box	Dilshad Garden
Building	137-B, Pocket-A
City	Delhi
State/Region	Delhi
Postcode	110095
Country	India
Telephone	+91-9886252062
E-mail	Shivani.garg@greneity.com
Website	www.greneity.com
Contact person	Shivani Garg
Title	Managing Director
Salutation	Ms.
Last name	Garg
Middle name	
First name	Shivani
Department	
Mobile	+91- 9886252062
Direct tel.	
Personal e-mail	Shivani.garg@greneity.com

APPENDIX 3- LUF ADDITIONAL INFORMATION

Risk of change to the Project Area during Project Certification Period:	
Risk of change to the Project activities during Project Certification Period:	
Land-use history and current status of Project Area:	
Socio-Economic history:	
Forest management applied (past and future)	
Forest characteristics (including main tree species planted)	
Main social impacts (risks and benefits)	
Main environmental impacts (risks and benefits)	
Financial structure	
Infrastructure (roads/houses etc):	
Water bodies:	
Sites with special significance for indigenous people and local communities - resulting from the Stakeholder Consultation:	
Where indigenous people and local communities are situated:	
Where indigenous people and local communities have legal rights, customary rights or sites with special cultural, ecological, economic, religious or spiritual significance:	

APPENDIX 4-SUMMARY OF APPROVED DESIGN CHANGES

Please refer to Design Change [Requirements](#) for more information on procedures governing Design Changes

Revision History

Version	Date	Remarks
1.2	14 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 yr rule) and Ongoing Financial Need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.1	24 August 2017	Updated to include section A.8 on 'gender sensitive' requirements
1.0	10 July 2017	Initial adoption